

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
3 HT	(a)			supply electricity to consumers (1) Meet demand / provide reliable supply (1)	2			2		
	(b)			X – (step up transformer) reduces current/increases voltage (1) reduces energy loss/more efficient transfer (1) Y – (step down transformer) reduces voltage (1) safe(r) levels (1)	4			4		
	(c)	(i)		Power for each shed = 230 × 80 (1) = 18 400 W(1) 100/18.4 (ecf) OR 100 000/18 400 (ecf)(1) = 5.44 – 5 buildings so yes it can power 5 sheds / total power needed is less than 100 kW (1) OR $\frac{100\text{ kW}}{5}$ (1) = 20 kW available for each shed (1) Power for each shed= 230 × 80 (1) = 18 400 W is less than the 20 000 available so can power the sheds(1) OR Current = $\frac{100\,000}{230}$ (1) =434.8 (1) $\frac{434.8}{5}$ (1) = 86.9 which is >80 A required for the shed so yes (1) OR		4		4	4	

				Maximum current for 5 sheds= 5×80 (1) = 400 A (1) Max voltage available = $\frac{100\,000}{400}$ (1) =250 V which is more than the 230 required so yes (1)						
		(ii)		Uses {sustainable/renewable} energy (1) CO ₂ released is absorbed by (new) plant growth (1) / Carbon neutral (1) So reduces contribution to Greenhouse effect (1)	3			3		
				Question 3 total	9	4	0	13	4	0